Cloud Computing

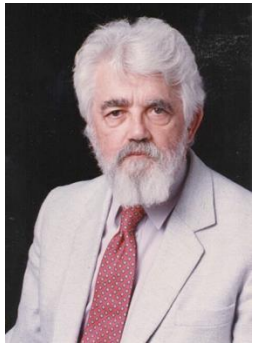
Lecture 1

Prof. Emiliano Casalicchio
Department of Computer Science
Sapienza University of Rome

Agenda

- Definition of cloud computing
  - NIST Special Publication 800-145
    - Essential characteristics
    - Service models
    - Deployment models
- Cloud Computing reference architecture (also flipped)
  - NIST SP 500-292
- Criteria for determining if a computing capability is offered as a cloud service (flipped)
  - NIST SP 500-322

A curiosity



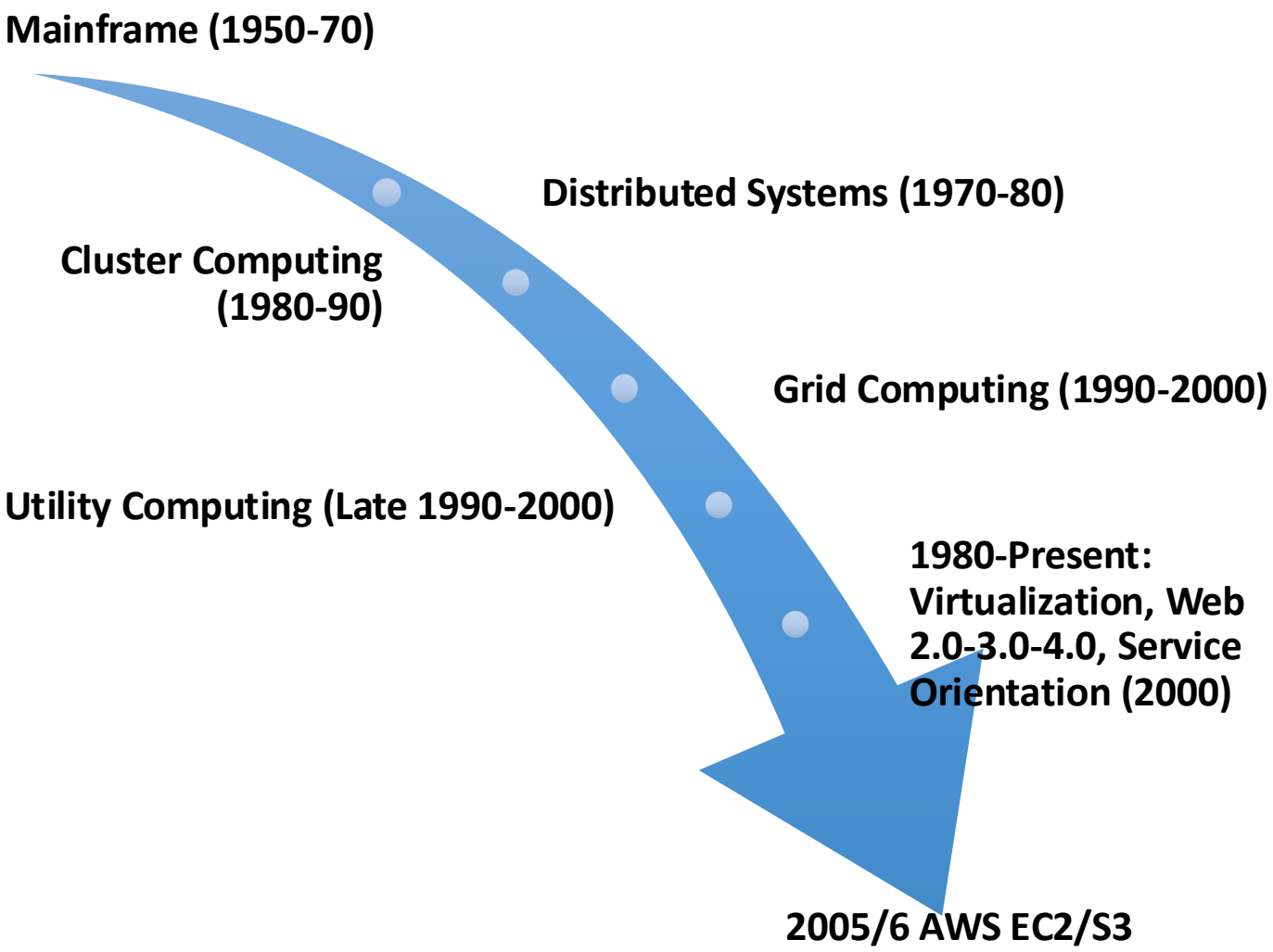
- 1961: John Mc Carthy - Standford Univ. – Inventor of LISP
  - *"If computers of the kind I have advocated become the computers of the future, then* ***computing may someday be organized as a public utility just as the telephone system is a public utility****... The computer utility* ***could become the basis of a new and important industry****."*

- 1969: Leonard Kleinrock, ARPANET project – Internet grandfather
  - "As of now, computer networks are still in their infancy, but as they grow up and become sophisticated, **we will probably see the spread of "computer utilities"**, which, like present electric and telephone utilities, will service individual homes and offices across the country."

Utility computing and Cloud Computing

- Today computing services are readily available on demand, just as other utility services
- Users (consumers)
  - pay providers only when they access the services
  - no investments or difficulties in building and maintaining complex IT infrastructure.
  - access services based on their requirements without regard to where the services are hosted
- This model has been referred to as *utility computing* or, since 2007, as *cloud computing*.

Evolution from a technological view point



History of cloud computing (cont'd)

- 2005/2006 Jeff Bezos, Amazon
- let users rent data storage and computer server time from Amazon like a utility
  - Elastic Compute Cloud (EC2)
  - Simple Storage Service (S3)

Examples of Cloud Services Today

Ca

NIST SP800-145: definition of cloud computing

- *Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction*

- *Cloud computing has essential characteristics …*

- *Cloud computing is based on a service model and on a delivery model*

Discussion

- Ubiquitous –server and data can be located anywhere and accessed from anywhere
  - Is it important or not where servers and data are located? Should we have control on that?
  - Does the location of the customer with respect to the location of the server/data could impact the perceived performance/quality of experience?

Essential characteristics (NIST)

- *Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.*
- *This cloud model is composed of five essential characteristics, three service models, and four deployment models*
- Essential characteristics
  - ***On-demand self-service***
  - ***Broad network access***
  - ***Resource pooling***
  - ***Rapid Elasticity***
  - ***Measured Services***
- Service models
  - *SaaS, PaaS, IaaS*
- Deployment models
  - *Private, Community, Public, Hybrid cloud*

- Introduce new concepts (elasticity, multitenancy, monitoring)
- Enabled by technologies (broadband, cluster computing, virtualization, …)

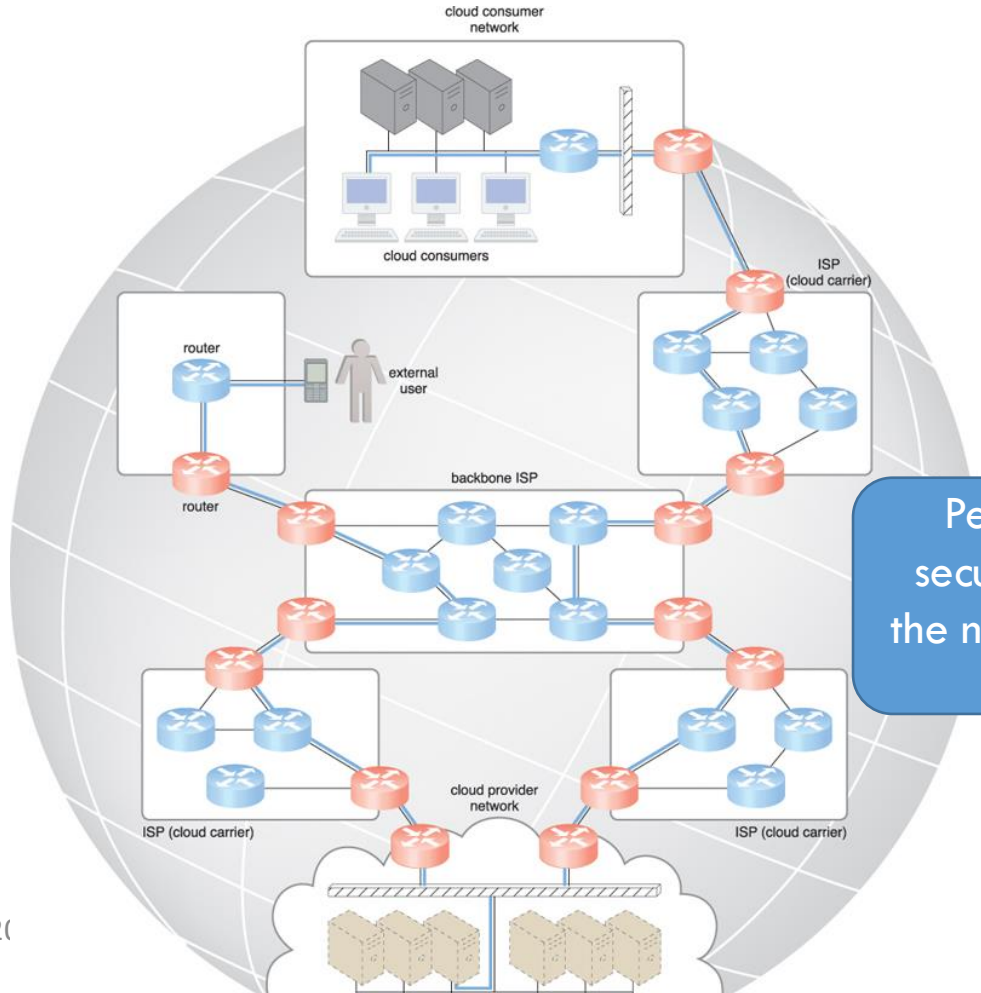
Level of control and separation of responsibility

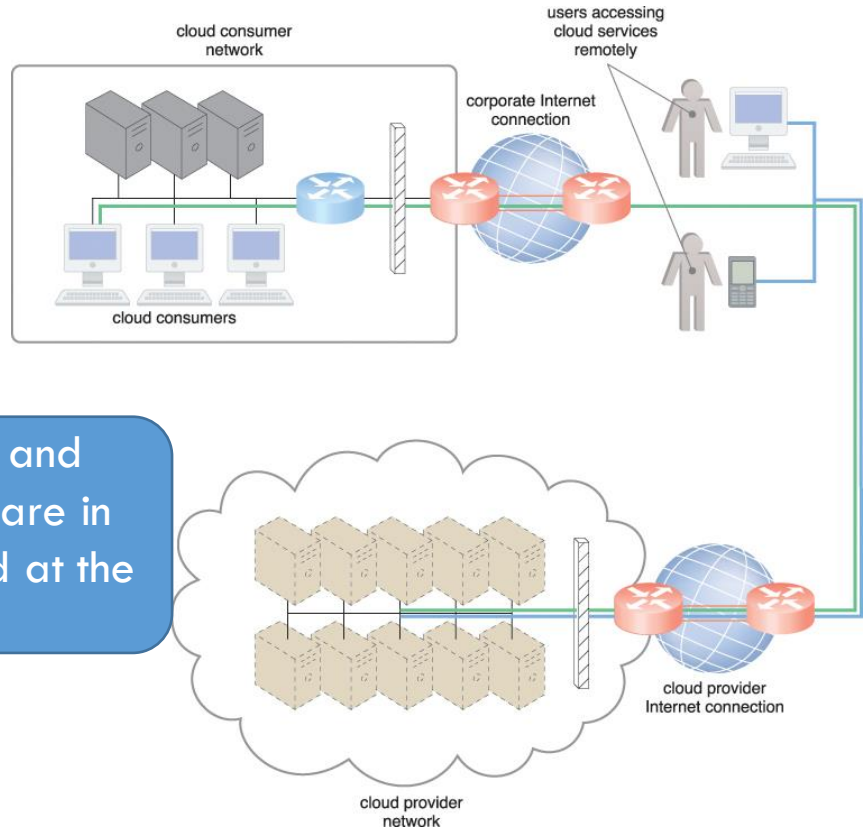
On-demand self-service

- Not required human intervention for resource provisioning
- Interfaces
  - Web interface (Openstack, AWS, Google Cloud)
  - Dedicated shell (e.g. AWS shell)
  - Programming API (e.g. AWS SDK for Java, Python, .Net, Ruby, Go, …)

Broad network access

- Refer to the possibility to access cloud resources from the internet using different devices
- Rely on broadband network and internet technologies (enabling technologies)



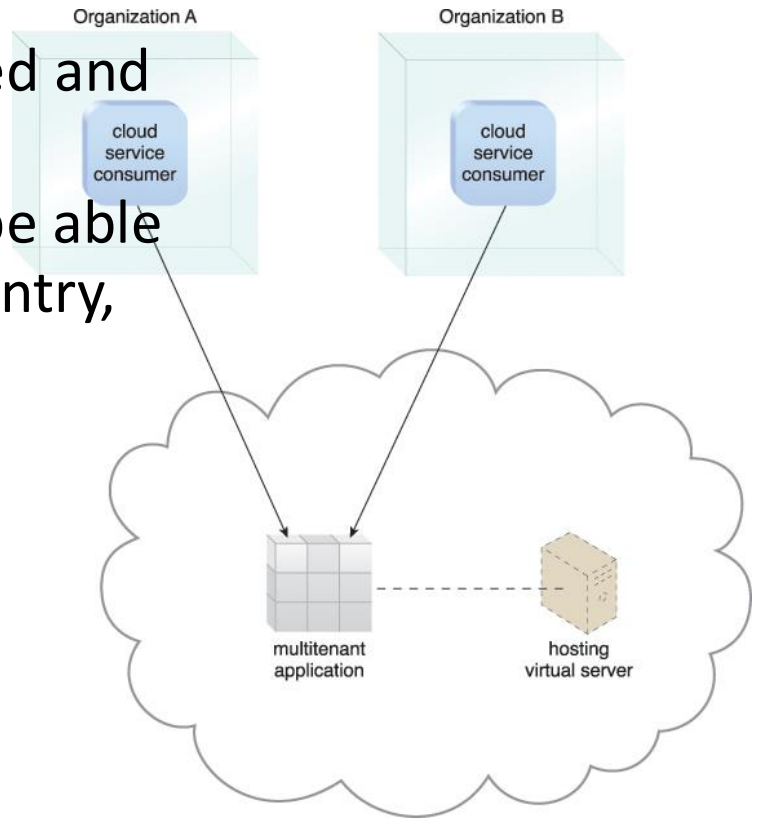


Performance and security issues are in the network and at the edge

salicchio@di.uniroma1.it

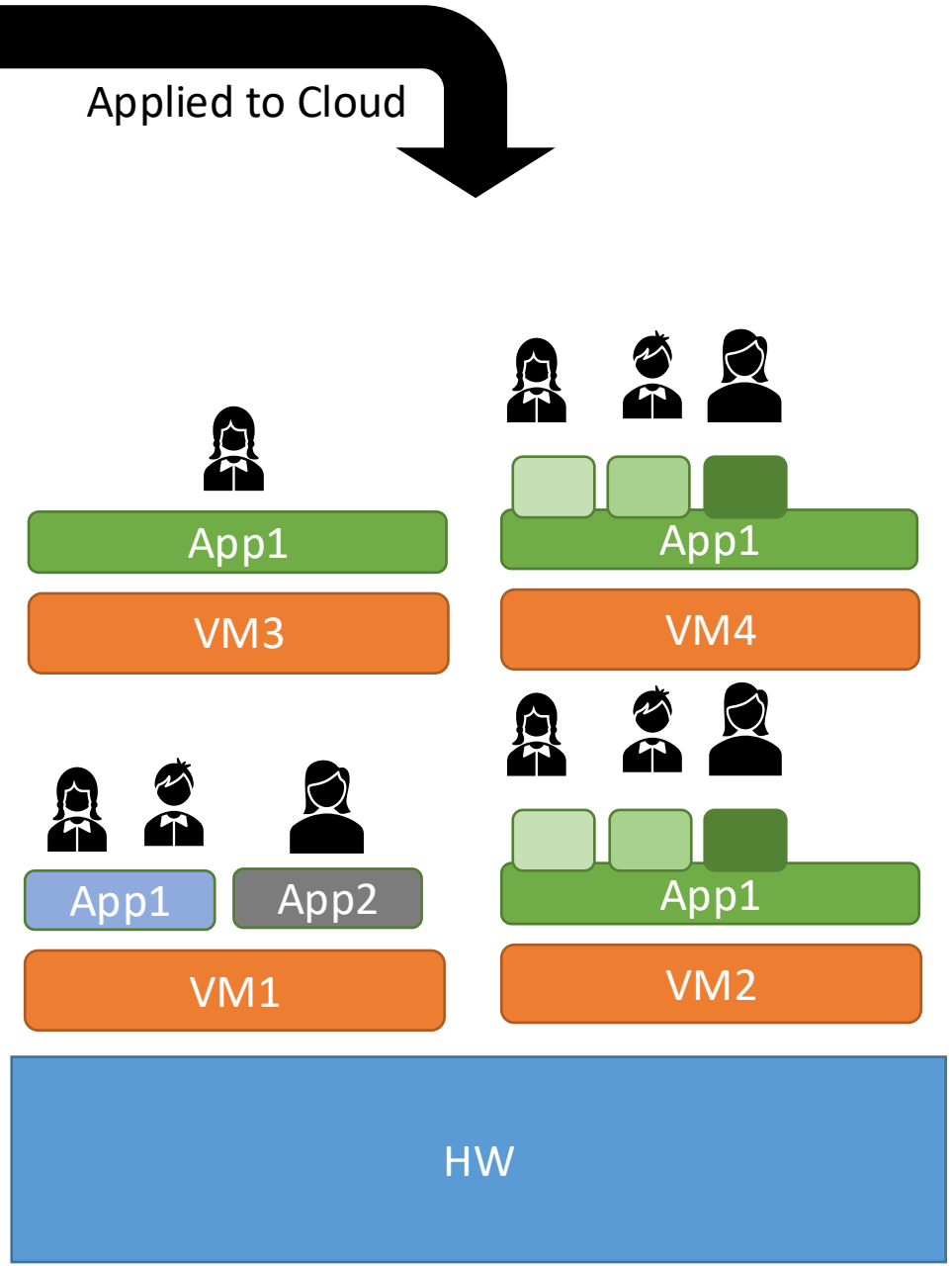
Resource pooling

- [NIST] Computing resources are pooled to serve multiple consumers using a **multi-tenant model**,
  - different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
  - location independence … no control or knowledge … may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter)



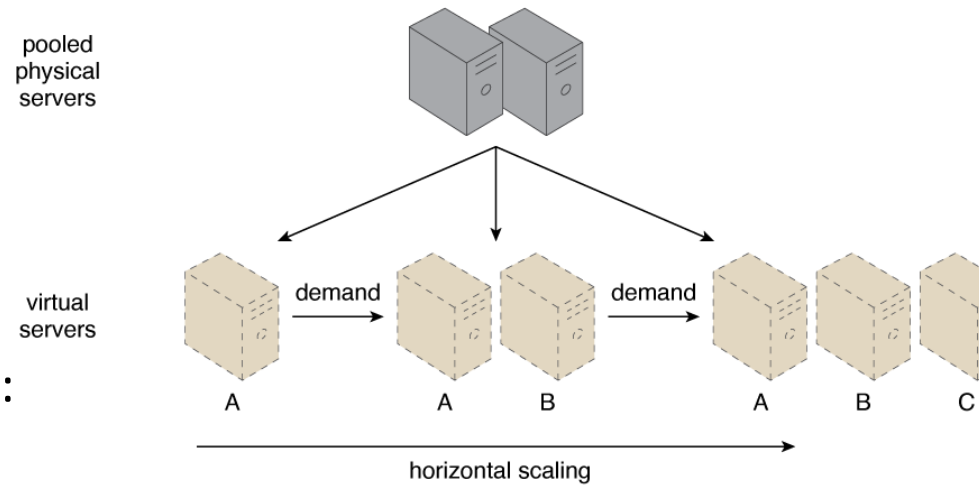
Multi tenant model

- Multi-tenancy (of applications)
  - multiple users (tenants) access the same application logic simultaneously.
  - Each tenant has its own view of the application as a dedicated instance and is unaware of other tenants
  - Multitenant applications isolate data and configuration information
- Multitenant application properties
  - *Usage Isolation* – performance and availability
  - *Data Security*
  - *Recovery from failures*
  - *Application Upgrades*
  - *Scalability*
  - *Metered Usage*
  - *Data Tier Isolation*

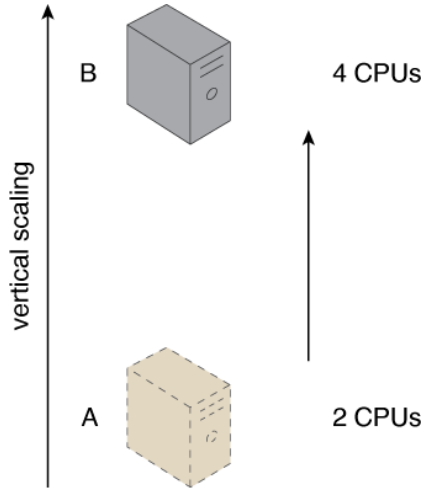


Elasticity

- Is the degree to which a system is able to adapt to workload changes by provisioning and deprovisioning resources in an automatic manner, such that at each point in time the available resources match the current demand as closely as possible [Elasticity in Cloud Computing: What It Is, and What It Is Not]
- Elasticity is related to the concept of Scalability
  - **Horizontal Scaling – scaling out and scaling in**
  - **Vertical Scaling – scaling up and scaling down**



Copyright © Arcitura Education



Copyright © Arcitura Education

Rapid elasticity

- This property allows to fulfill the service demand; E.g.
  - an e-commerce web site on the black-friday week
  - A newspaper web site in case of a breaking news
  - Large scale events like Olympic games, world champions, and so on …
  - Entertainment applications: online games, streaming of music and movies, …

Rapid Elasticity

- Let us consider a IaaS provider, AWS
  - (All this concepts will be explained in detail later)

- With the EC2 service you can create VMs manually
  - You can manually add VMs (EC2 instance), that is you can scale / you can’t be rapid in following the demand

- You need some automatism, e.g.
  - monitor VMs CPU usage (continuously)
  - if CPU usage > 70% for 1 min. add 1 VM
    - When you have more VMs you can take decision on the average CPU usage

- That is the AWS auto-scaling and elastic load balancing

Measured service

- Cloud provider charge customers for
  - h/min/sec a VM or other virtualized resource is used
  - MB of storage used / MB of I/O transfer
  - MB/GB transferred over the network
  - …
- Cloud providers should guarantee specific Service Level Agreement (SLA)
  - 99.9% up time
  - A give bandwidth
  - A certain amount of CPU share
  - …
- That requires measurement/monitoring capabilities for
  - Proper billing
  - SLA management

t2.small	1 CPU	Variable ECU	2 GB RAM	$0.023 per Hour
m4.large	2 CPU	6.5 ECU	8 GB RAM	$0.1 per Hour

Monthly Uptime Percentage	Service Credit Percentage
Less than 99.95% but equal to or greater than 99.0%	10%

Measured services (cont'd)

- Consumers need to monitor the usage of their resources to automate management tasks, e.g
  - Scaling (autoscaling thresholds)
  - Controlling costs (alerts)
  - Guarantee application specific SLA (throughout, response time, …)
  - **Increase resiliency**

Measured services (cont'd)

- Resiliency
  - *the ability to provide and maintain an acceptable level of service in the face of faults and challenges to normal operation*
- E.g. (this concepts will be explained in detail later)
  - AWS Elastic load balancer
    - If a node is down automatically is excluded by the pool
  - AWS Autoscaling
    - You can configure the minimum number of servers you want always up despite failures
  - Docker container orchestration (e.g. Kubernetes, Swarm)
    - You can define the desired number of container always up and running despite failures

That require measurable services and monitoring

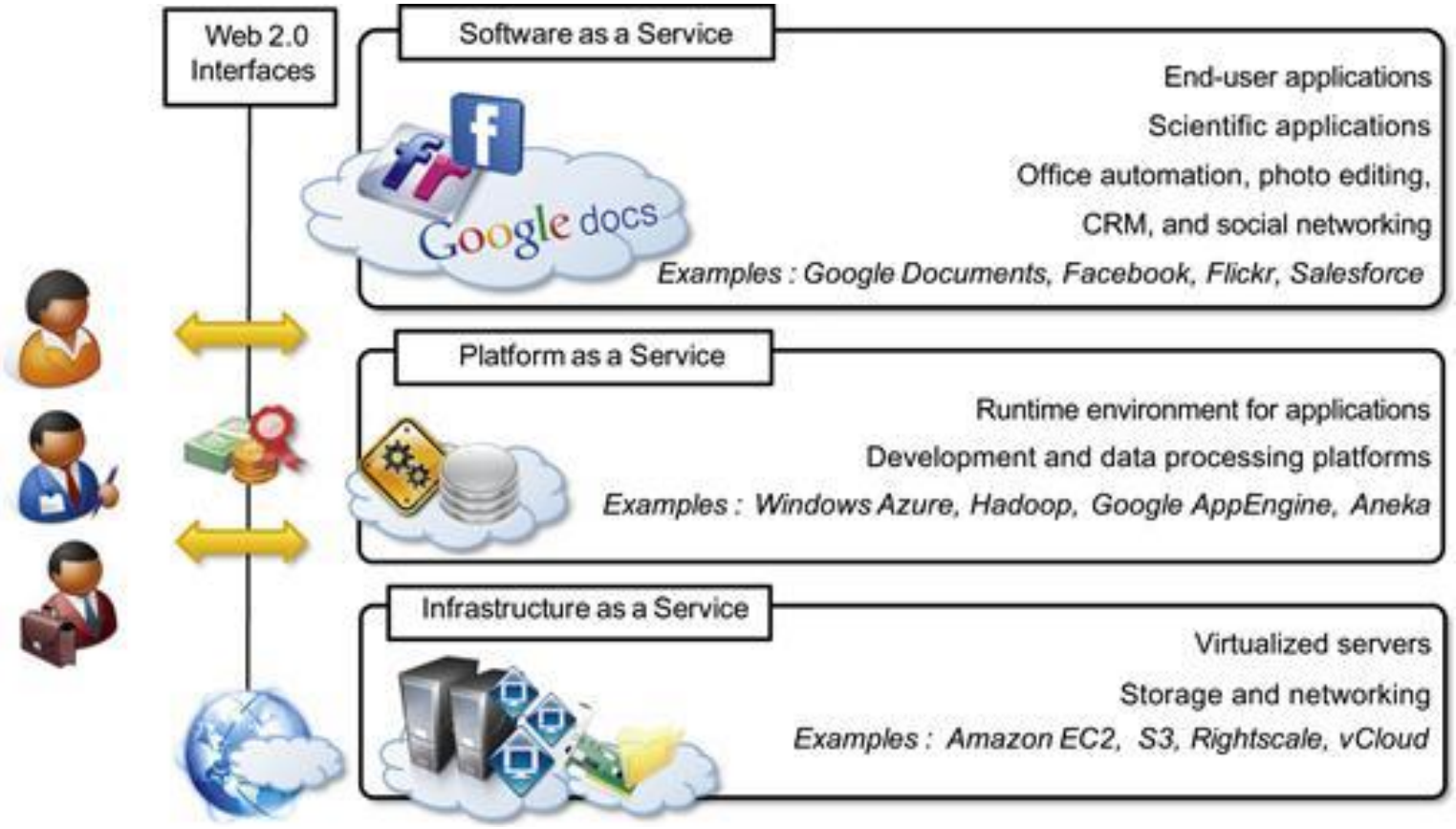
Definition of cloud computing (NIST)

- *Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.*
- *This cloud model is composed of five essential characteristics, three service models, and four deployment models*
- Essential characteristics
  - *On-demand self-service*
  - *Broad network access*
  - *Resource pooling*
  - *Rapid Elasticity*
  - *Measured Services*
- Service models
  - *SaaS, PaaS, IaaS*
- Deployment models
  - *Private, Community, Public, Hybrid cloud*

- Introduce new concepts (elasticity, multitenancy, monitoring)
- Enabled by technologies (broadband, cluster computing, virtualization, …)

Level of control and separation of responsibility

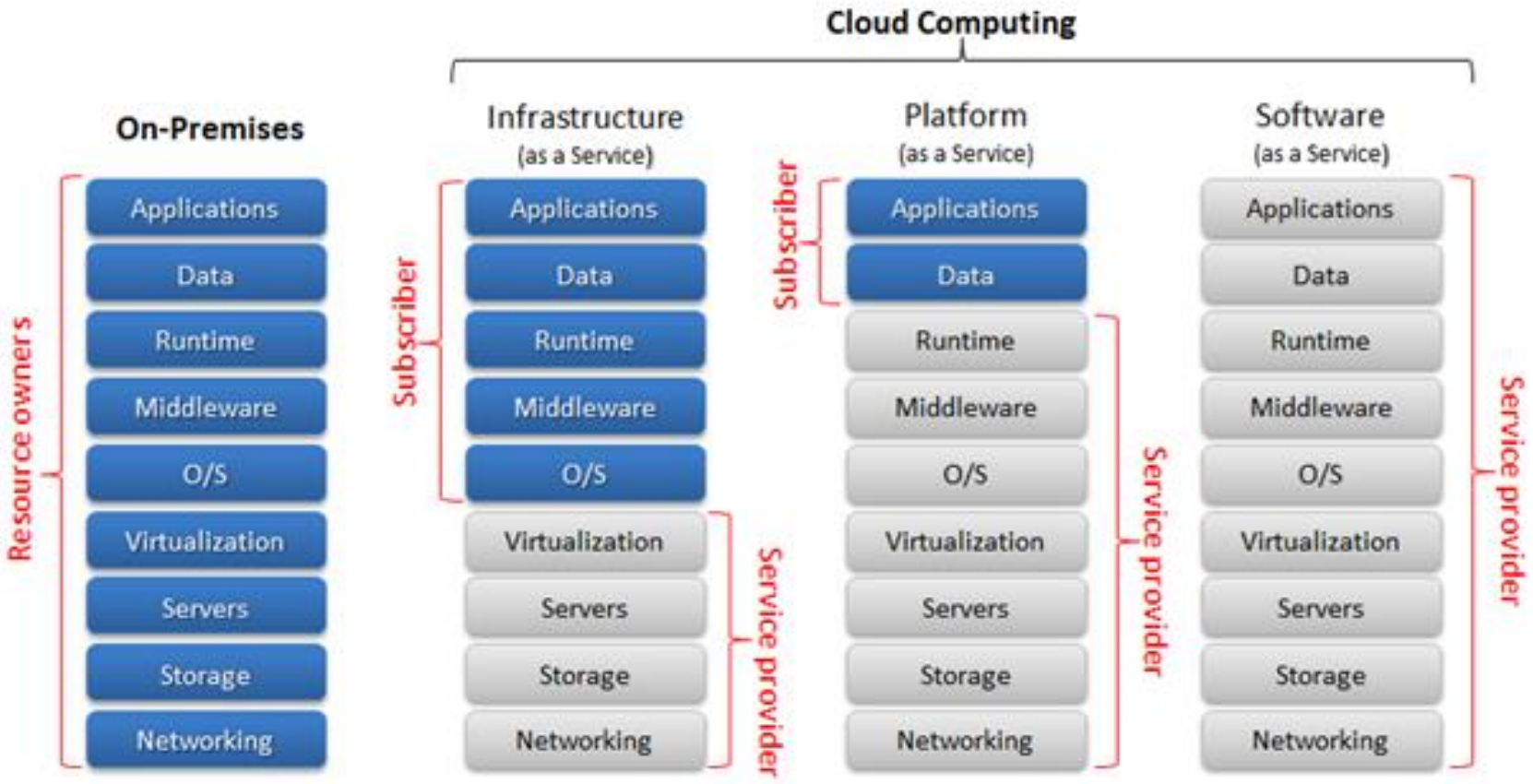
Service model



- The consumer can control only user-specific application configuration settings.
- The consumer has control over the deployed applications and possibly configuration settings for the application-hosting environment
- The consumer has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components

Service models, Level of control and separation of responsibilities

- Limit the control of the cloud consumer (subscriber)
- Separate the resposabilities of the cloud consumer and cloud provider (service provider)
- Moving from IaaS to SaaS
  - the control of the consumer on the resources decrease
  - the expertise needed to manage the cloud service decrease
  - the easy of use increase

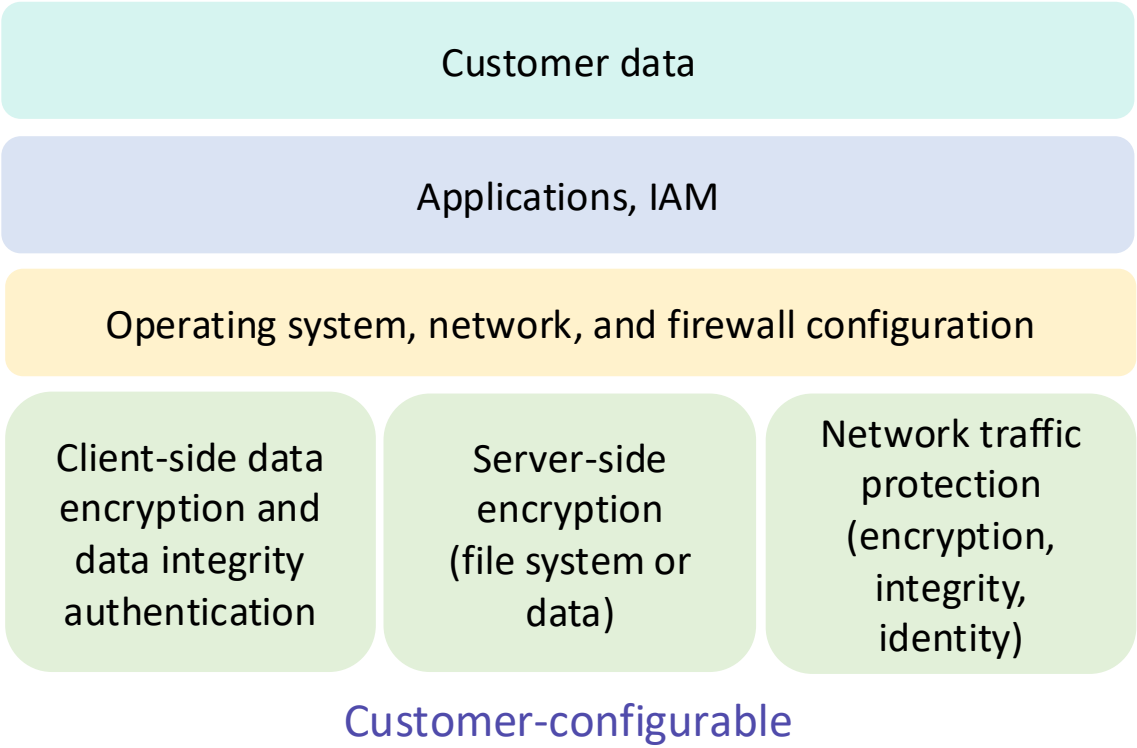


Source: https://blogs.technet.microsoft.com/yungchou/2010/11/15/cloud-computing-primer-for-it-pros/

Security *of* the cloud - Cloud provider responsibilities

- Physical security of data centers
  - Controlled, need-based access
- Hardware and software infrastructure
  - Storage decommissioning, host operating system (OS) access logging, and auditing
- Network infrastructure
  - Intrusion detection
- Virtualization infrastructure
  - Instance isolation

Customer responsibility: Security *in* the cloud



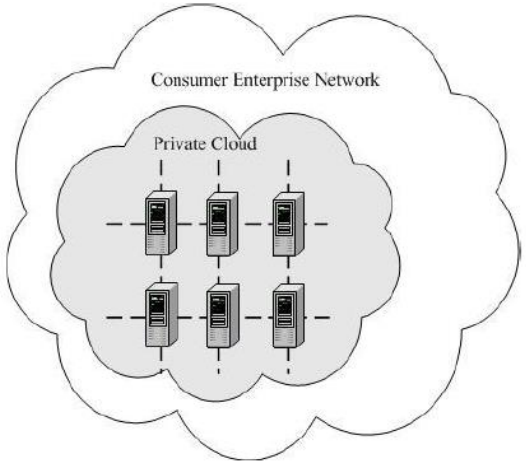
- Virtual machines instance operating system
  - Including patching, maintenance
- Applications
  - Passwords, role-based access, etc.
- OS or host-based firewalls
  - Including intrusion detection or prevention systems
- Network configurations
- Account management
  - Login and permission settings for each user

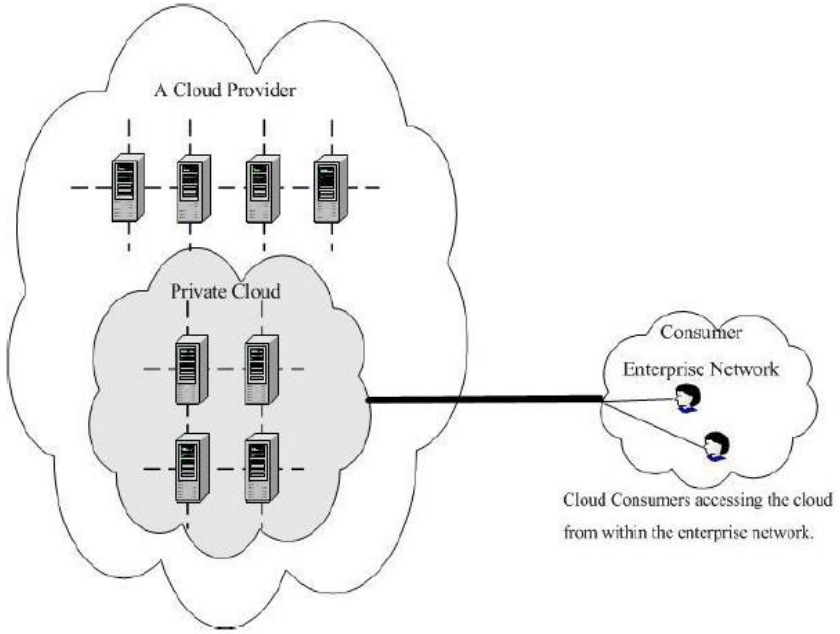
Deployment models: Public, Private and Hybrid cloud

- Could you make an example of public cloud?
- Could you make an example of privite cloud?
- Could you make an example of hybrid cloud?

Deployment models

- Public cloud
  - All you can access!!!!
- Private cloud
  - Any cloud dedicated to a specific customer and in logical/physical isolation with other customers
  - On-premise
  - Virtual Private Cloud
    - Amazon VPC and Goodle Cloud VPC focus on providing VPN feature to connect your resources
    - IBM, HP clouds gives more physical isolation
- Hybrid





Criteria for determining if a computing capability is offered as a cloud service

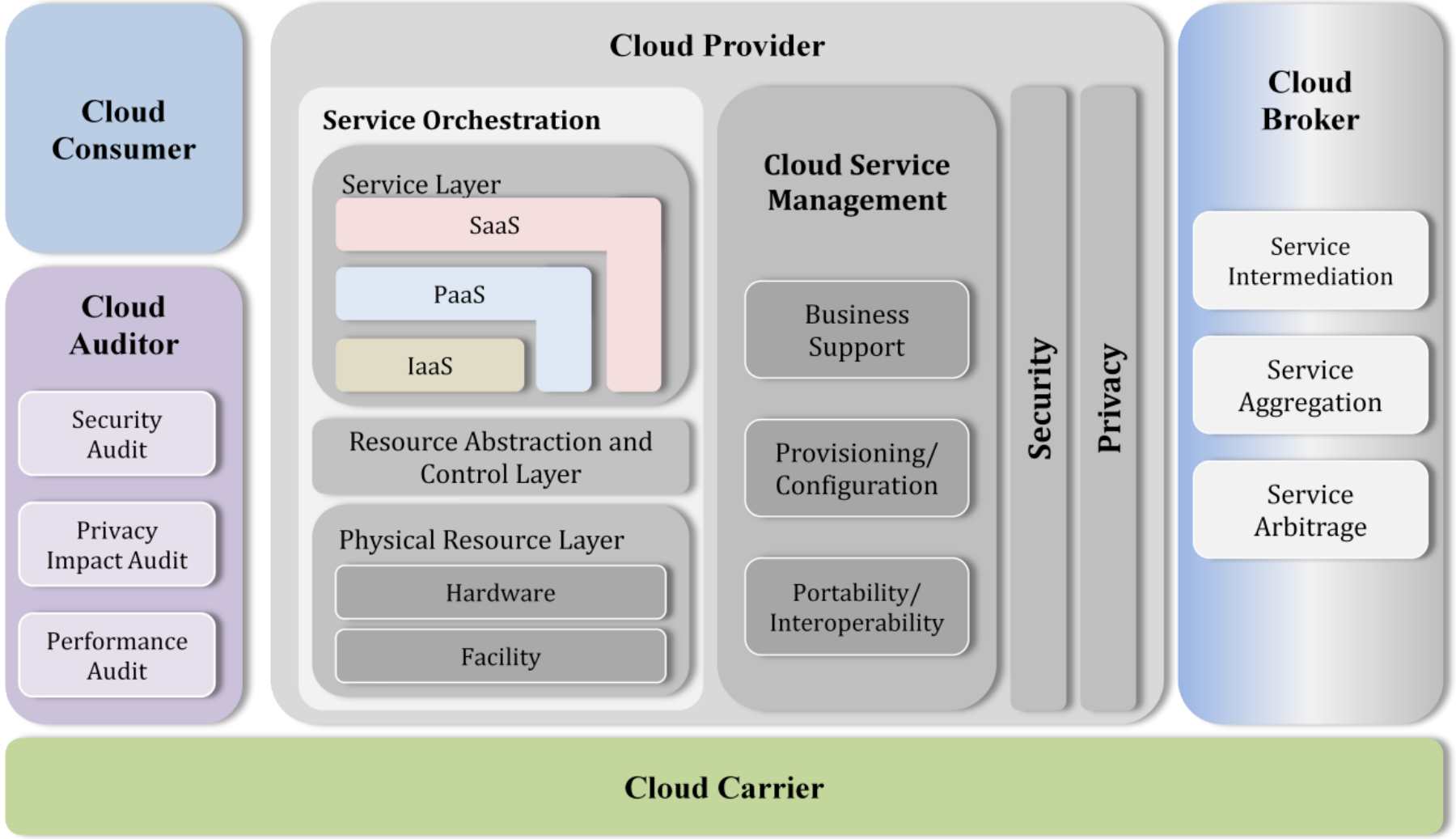
NIST SP 500-322

Casalicchio@di.uniroma1.it

Evaluation of Cloud Computing Services Based on NIST 800-145

- Used to determine whether a service is a cloud service
- Allow to verify
  - If a service is a cloud service (Cloud Service Worksheet)
  - What service model is used (Cloud Service Model Worksheet)
  - What deployment model is used (Cloud Deployment Model Worksheet)

NIST SP 500-292:
The Cloud computing NIST Reference Model

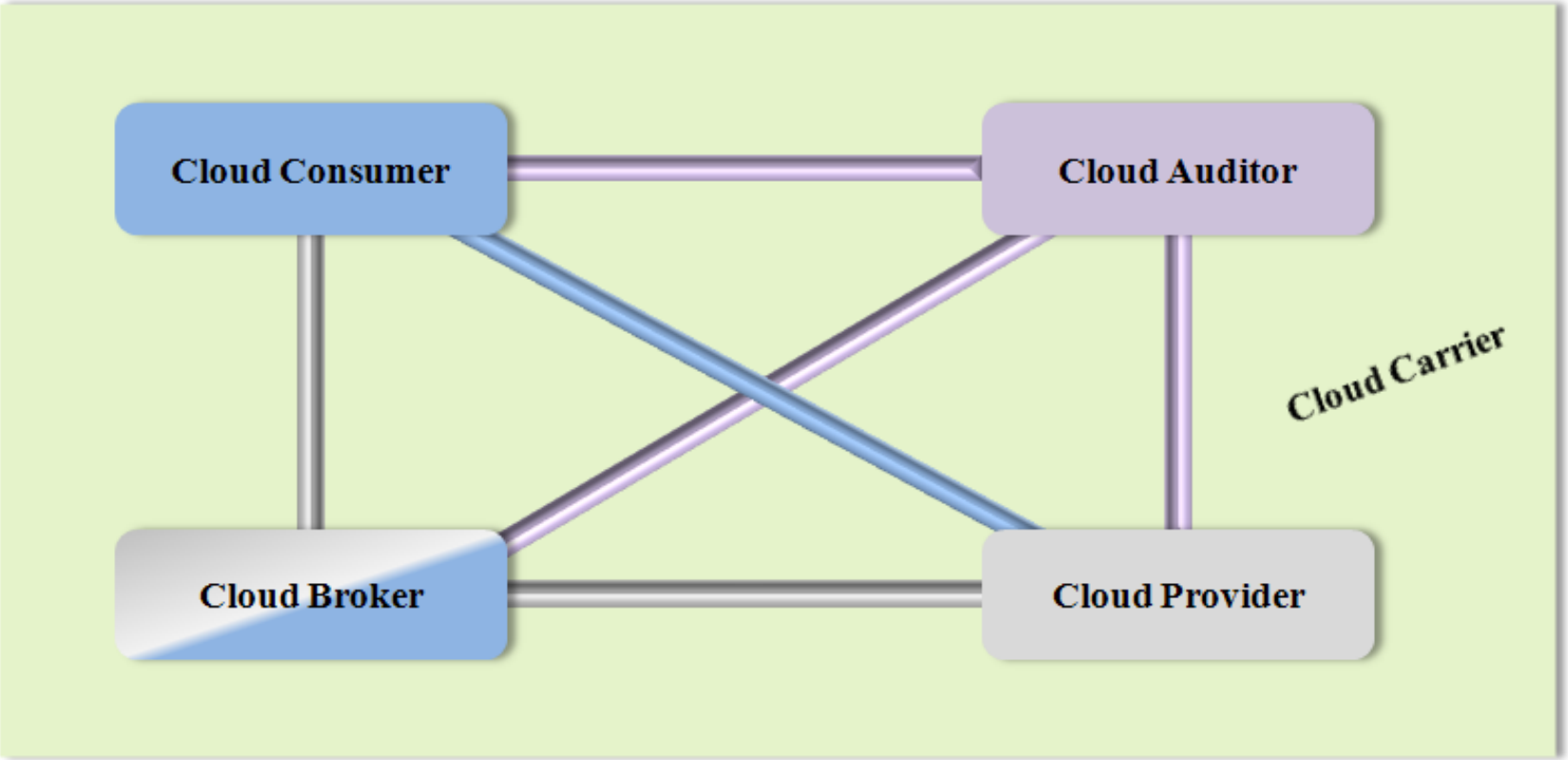


Casalicchio@di.uniroma1.it

Cloud roles

Definition	
A person or organization that maintains a business relationship with, and uses service from, *Cloud Providers*.	Cloud Consumer
A person, organization, or entity responsible for making a service available to interested parties.	Cloud Provider
A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.	Cloud Auditor
An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between *Cloud Providers* and *Cloud Consumers*.	Cloud Broker
An intermediary that provides connectivity and transport of cloud services from *Cloud Providers* to *Cloud Consumers*.	Cloud Carrier

Interactions between the Actors in Cloud Computing



The communication path between a cloud provider and a cloud consumer
The communication paths for a cloud auditor to collect auditing information
The communication paths for a cloud broker to provide service to a cloud consumer

Examples

Cloud Consumer
Cloud Broker
Cloud Provider 1
Cloud Provider 2

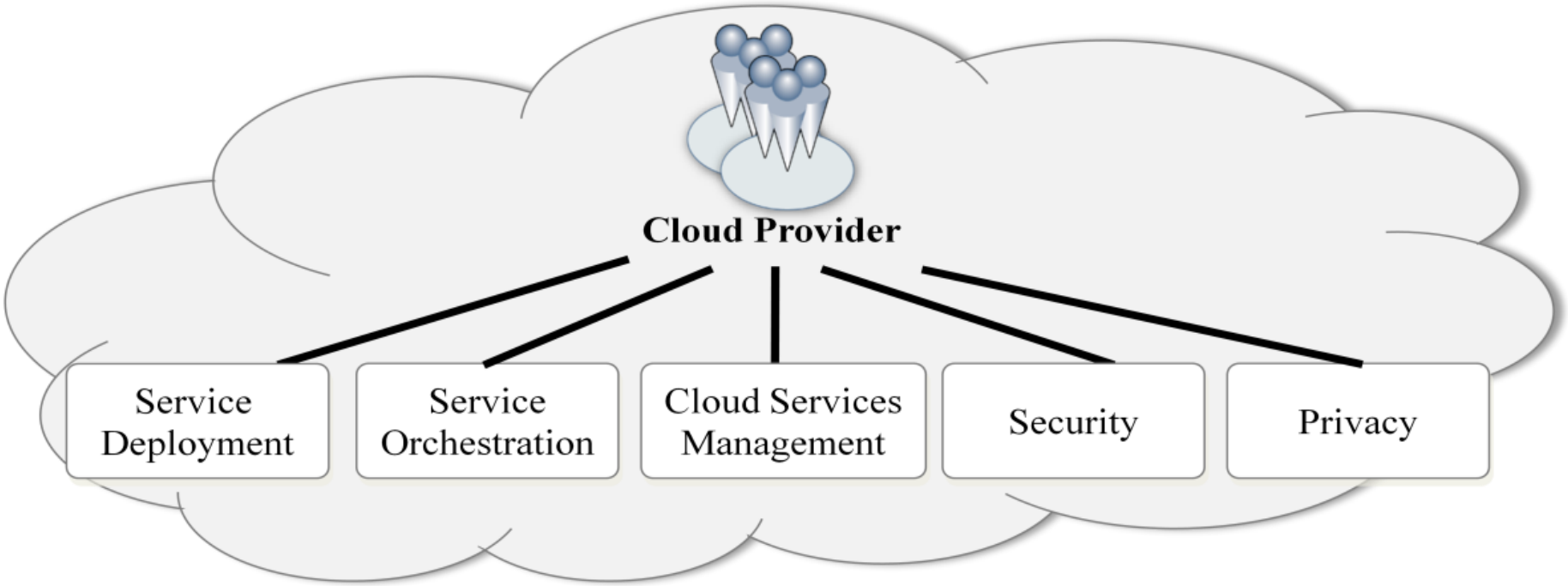
Cloud Consumer
SLA1
Cloud Provider
SLA2
Cloud Carrier

Cloud Consumer
Cloud Provider
Cloud Auditor

Cloud Broker

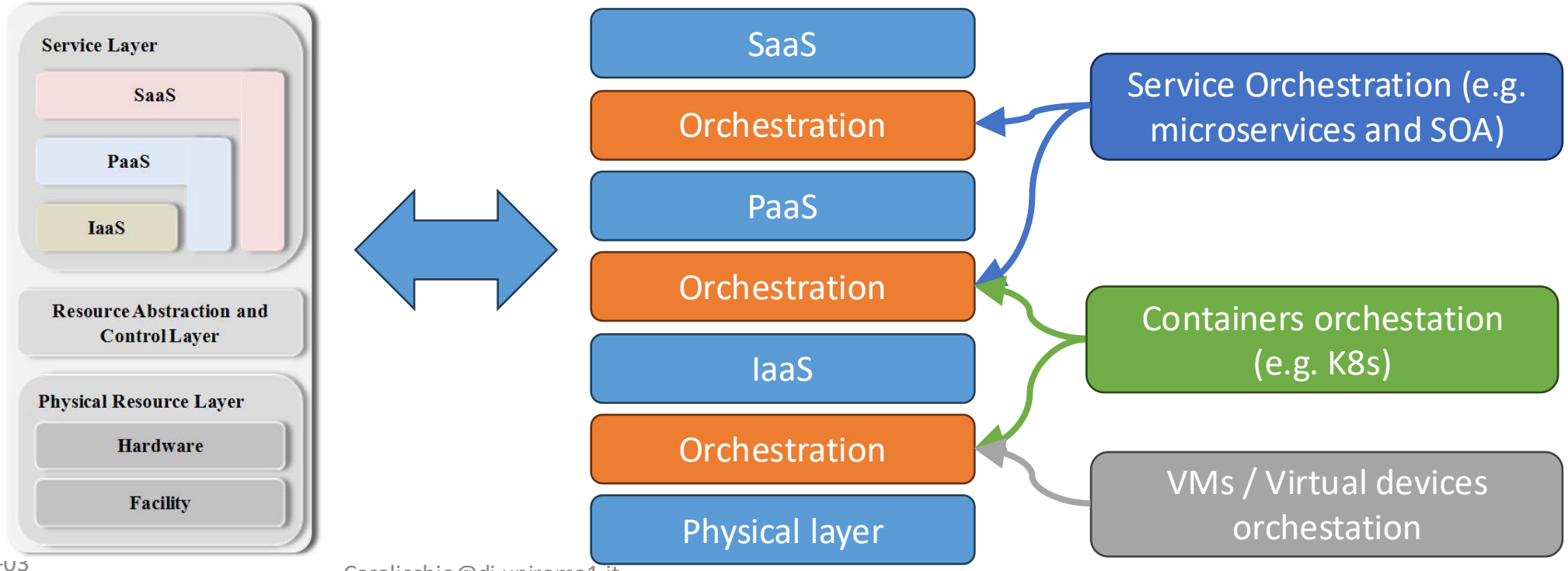
- Service Intermediation
  - Improving specific capabilities
  - Providing added value services
- Service Aggregation
  - Combine and integrate to implement a new service
- Service Arbitrage
  - Dynamic Aggregation

CSP activities



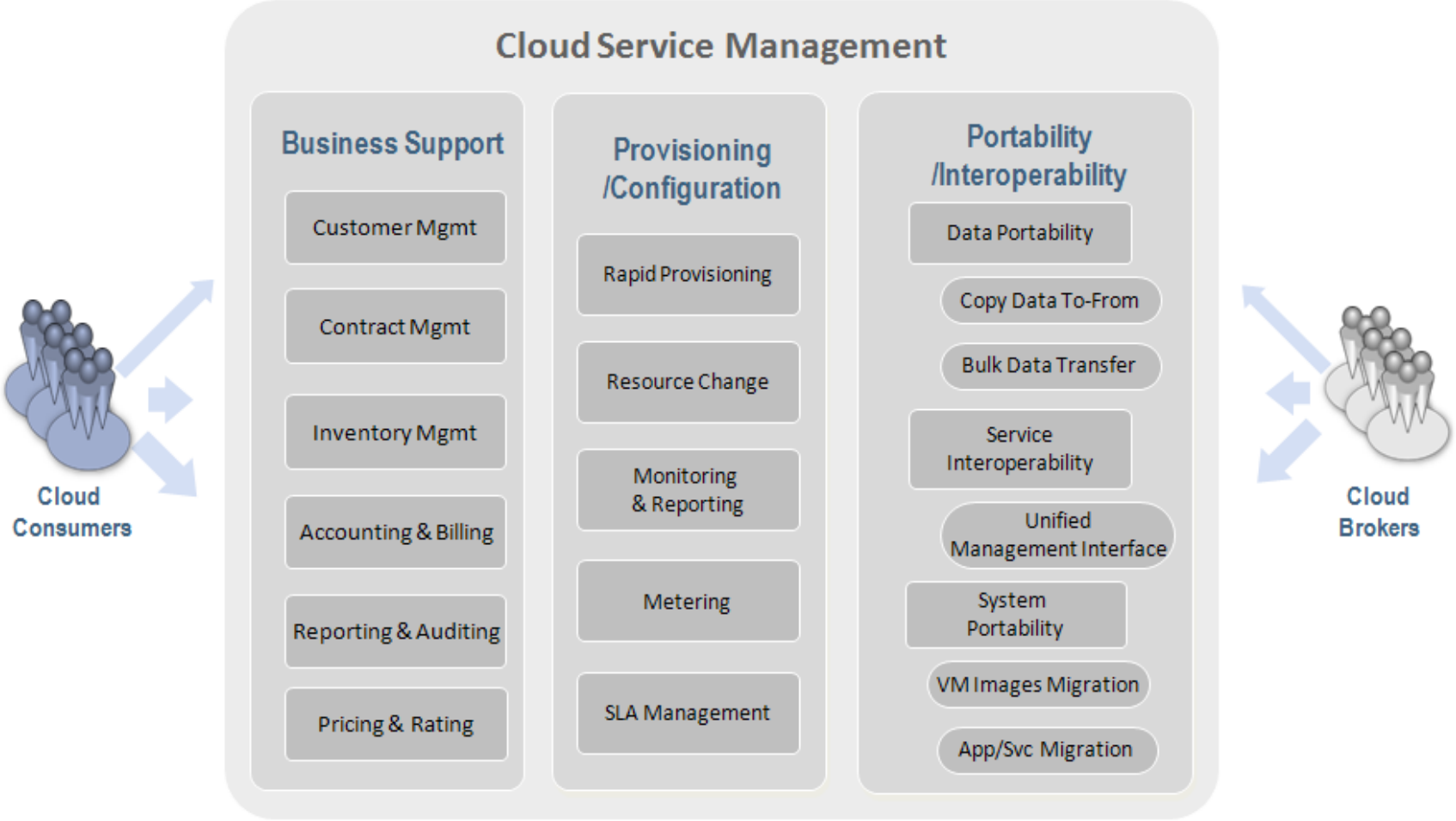
Service Orchestration

- *Cloud Service Orchestration* refers to the composition of system components to support the Cloud Providers activities in arrangement, coordination and management of computing resources in order to provide cloud services to Cloud Consumers



Cloud Service Management

- *Cloud Service Management* includes all of the service-related functions that are necessary for the management and operation of those services required by or proposed to cloud consumers.



Summary

- Cloud computing is an implementation of the concept of utility computing
  - *Cloud computing is a model for enabling ubiquitous, convenient,* ***on-demand network access*** *to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction*

Summary cont'd

- Essential characteristics
  - On-demand self-service
  - Broad network access
  - Resource pooling
  - Rapid Elasticity
  - Measured Services

- Deployment and service models

Learning Outcome

- You should be aware of, and capable to explain
  - Concept of utility computing
  - Drivers for cloud computing
  - What is cloud computing,
  - The main characteristics,
  - Cloud service and deployment models
  - Cloud reference architecture
    - Roles on cloud computing

Assessment: Analysis, Knowledge and understanding, Critical thinking, Language

This knowledge will be assessed in the oral exam

Questions?

Casalicchio@di.uniroma1.it